



LIGHT WAVE DEHAZE

Team Members

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Problem Statement

$$I(x) = J(x)t(x) + A\big[1 - t(x)\big]$$



Hazy image I



Clean image J

- Recover a clean image from a single hazy observation.

Why Dehazing Is Needed

Haze obscures distant objects



Preprocessing for detection and tracking



Existing Dehazing Approaches

Approach	Core idea	Key consideration
Physical Prior	Restore images using assumptions about haze and the scene	Depend on how well the assumptions match the scene
CNN	Learn features through convolution	Ability to capture wider context
Transformer	Model relationships across image regions	Computational and memory costs
Diffusion	Restore images through denoising	Latency with multiple inference steps

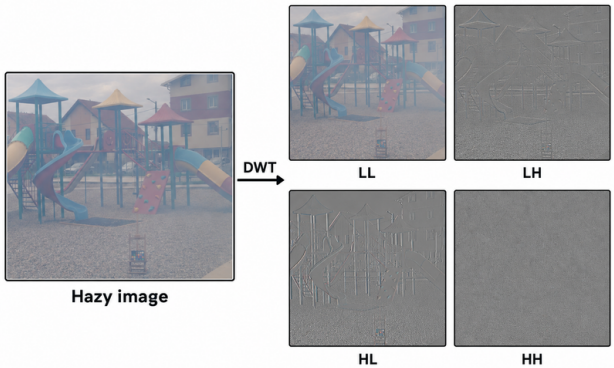
Trade-offs depend on the architecture and its optimization.

Combining Strengths

How can we use the strengths of each method at different stages of image processing?



Frequency-Domain Observation

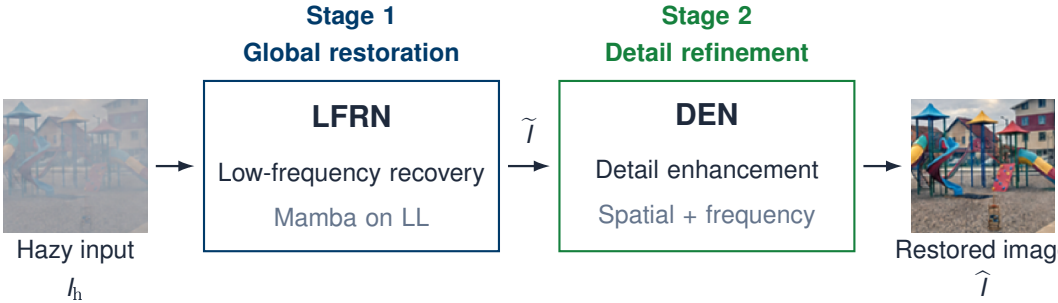


Low-frequency component (LL)
Global appearance and scene structure.

High-frequency components
Edges and local details in LH, HL, and HH.

Key idea
Match each method's strengths to its restoration stage.

Overall Design and Objectives



- **Restoration:** remove global haze while preserving local details.
- **Efficiency:** balance restoration quality with inference cost.

Evaluation Metrics

Metric	What it measures	How to interpret it
PSNR (dB)	Peak signal-to-noise ratio Pixel accuracy against a clean reference.	Higher values indicate lower pixel error.
SSIM	Structural similarity index Compares structure, contrast, and luminance with a clean reference.	Values closer to 1 indicate greater similarity.
FPS	Frames per second Number of images processed per second.	Higher values indicate faster processing.

Compare FPS using the same hardware, image resolution, and batch size.

Further Optimization Targets

Targets on the SOTS dataset

$\geq$ 25 dB	$\geq$ 0.95	$\geq$ 20
PSNR	SSIM	FPS

- Optimize the frequency-domain and RGB restoration blocks.
- Assess latency, parameters, and FLOPs for future edge deployment.

Proposed targets, not reported results. Specify the hardware and image resolution for FPS.

Thank You

Questions and Discussion